



DESIGN NOTES

- Input: 3.8V-4.2V DC (Single-Cell Li-Ion)
- Output: 3.3V DC, 500mA Max
- LDO: LP5912-3.3DRVR
- Reverse Polarity Protection: Q1 (P-MOSFET)
- Overcurrent Protection: F1 (PTC Fuse)
- Transient Protection: D1 (TVS Diode)
- Gate Protection: D2 (Zener Diode)
- Power Indicator: Green LED on VOUT
- Test Points: VIN (TP1), VOUT (TP2), GND (TP3)
- Operating Temperature: 0°C to 70°C

Designed by Nikhil Abboju
Input: 3.8V-4.2V Li-Ion Battery
Output: 3.3V @ 500mA Max

Sheet: /
File: Lilon_3V3_500mA_Regulator.kicad_sch

Title: Single-Cell Li-Ion 3.3V 500mA Regulator Board

Size: A4 Date: 2026-06-09

KiCad E.D.A. 10.0.3

Rev: REV1.0

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